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EDUCATION

Purdue University

Master of Science in Electrical and Computer Engineering

West Lafayette, IN

January 2026 – December 2027

Relevant Coursework: Solid State Devices, Wireless Communication Networks, Probability.

University of Mumbai, Ramrao Adik Institute of Technology

Mumbai, MH

Bachelor of Engineering in Electronics and Telecommunications | GPA: 9.18/10 | Dept. Rank 4

June 2023

Relevant Coursework: Engineering Mathematics, Microcontrollers, Electromagnetics and Antenna, Microwave Engineering, Optical Communication Networks, Artificial Neural Networks, Data Structures and Algorithms, Project Management.

RESEARCH EXPERIENCE

Microwave Engineering (MWE) Lab, Ramrao Adik Institute of Technology

Mumbai, MH

Research Assistant

August 2022 – November 2022

- Optimized metamaterial unit cells for microstrip impedance matching at 2.4 GHz, achieving a 15% bandwidth enhancement.
- Designed split-ring resonator combinations using CST Studio Suite, supported by mathematical modeling of loading effects.
- Fabricated and tested FR-4 PCB prototypes; validated results via VNA measurements and on-site studies at Vedang Radio Technologies, Talaja.
- Simplified design equations to improve accessibility and practical implementation.

Antenna Design Lab, Ramrao Adik Institute of Technology

Mumbai, MH

Research Assistant

January 2022 – April 2022

- Designed a compact dual-band microstrip patch array for 5G at 3.5/5.5 GHz, achieving >20 dB isolation.
- Applied innovative DGS patterns to enhance isolation and ensure dual-band functionality.
- Developed cost-effective FR-4 prototypes, validated through HFSS simulations and empirical testing.
- Simplified antenna design processes by optimizing DGS geometry.

PROJECTS

Landmine Detection Rover with Real-Time GPS Mapping

RAIT, Mumbai

Project Leader

August 2022 – April 2023

- Designed and developed a versatile landmine detection system focused on safety and economic feasibility for impoverished regions.
- Built an adaptable prototype with GPS (NEO-6M), self-navigation, and mapping; detected coordinates were transmitted via the GPS module to a Google Sheet with Google Maps pin locations of the mines.
- Implemented Wi-Fi (ESP8266) for remote control and integrated HC-SR04 ultrasonic sensors for obstacle avoidance.
- Used lightweight, structurally rigid materials (MDF wood) for the chassis; powered by an ARM-based microcontroller with a metal detector and warning buzzer for landmine detection.

PWM (Pulse Width Modulation) Generator using Verilog on Xilinx FPGA

RAIT, Mumbai

Project Leader

January 2022 – April 2022

- Developed a PWM generator in Verilog on a Xilinx FPGA to efficiently control electrical signal power, specifically for LED intensity modulation.
- Advanced project scope from basic PWM concepts to controlling multiple LEDs, demonstrating progressive application of Verilog on the Xilinx simulator and FPGA boards.

PUBLICATIONS

Landmine Detection Rover with Automated Area Scanning and Real-Time GPS Mapping

June 2023

DOI: [10.13140/RG.2.2.23205.31208](https://doi.org/10.13140/RG.2.2.23205.31208)

- Integrated GPS localization, metal detection, and obstacle avoidance; transmitted real-time location data to Google Sheets via Wi-Fi. Powered by dual interconnected Arduino Nano microcontrollers, the system features a metal detection coil, proximity sensors, and a GSM-based alert system to notify personnel of detected mines. The lightweight prototype ensures mobility across challenging terrain.

WORK EXPERIENCE

Tidal Wave Technologies Pvt. Ltd.

Mumbai, India

Networking and RF Engineer

July 2023 – September 2025

- For the n78 radio used in the company's *Network-in-a-Box* (a portable, software-based Private 5G gNB for tactical and emergency-response use), designed and simulated microstrip patch antenna (MSPA) elements in CST Studio Suite, iterating on patch geometry and feed dimensions to improve return loss and impedance bandwidth across 3.3 to 3.8 GHz; the chosen design was carried forward to FR-4 prototyping and VNA characterization.
- To validate prototype RF boards against 3GPP Rel-15 limits, ran routine lab measurements (return loss, EVM, ACLR, spectral mask) on a VNA and signal analyzer, logged results across builds, and flagged regressions to the senior RF team, enabling faster turnaround on design fixes.
- With end-to-end test cycles bottlenecked by manual iperf and ping runs on the SDR-based gNB and open-source 5G core, wrote Python scripts to automate throughput, latency, and packet-loss sweeps, cutting per-cycle test time substantially and freeing senior engineers from repetitive bring-up work.
- To validate the kit's rapid-deploy use case in real conditions, supported outdoor field trials with commercial 5G phones as UEs by setting up the network on-site and capturing KPI logs, which helped senior engineers debug coverage and handover issues observed in the field.

Vedant Radio Technologies Pvt. Ltd.

Mumbai, India

Intern

May 2022 – August 2022

- Engineered RF front-end subsystems for LTE and 5G cellular networks, focusing on Doherty Power Amplifiers with optimized PAPR in 2T2R and 4T4R configurations.
- Collaborated on power amplifier and signal processing projects for next-generation telecommunications infrastructure.

GeeksForGeeks

India

Technical Content Writer Intern

October 2020 – April 2021

- Conceptualized and published detailed technical articles and blog posts on emerging technologies in AI, electronics engineering, and industry trends, driving a 30% increase in social media engagement.
- Produced high-quality blog posts covering machine learning, cybersecurity, and electronic systems, enhancing the platform's knowledge repository.

ACTIVITIES AND LEADERSHIP ROLES

Lead Member, The Wall & Social Wing, RAIT

September 2022 – June 2023

- Wrote, edited, compiled, and evaluated articles for the Annual Collegiate Literary Magazine.
- Organized 2 Open Mic events and secured sponsorship worth INR 50,000 from local businesses around the university.
- Taught underprivileged children for two weeks at Girija Orphanage, Kharghar, in collaboration with the Social Wing of RAIT.

Lead Curator, TEDxDYPatilUniversity

March 2021 – December 2022

- Spearheaded the Curation and Marketing Team for TEDxDYPatilUniversity, overseeing the selection of impactful speakers and content.

SKILLS & INTERESTS

Technical: C, C++, SQL, Excel, MATLAB.

Languages: English (Expert Proficiency), Hindi (Expert Proficiency), Spanish (Beginner Proficiency).

Laboratory: CST Studio Suite, Vector Network Analyzer (VNA), Microcontroller Programming.

Interests: Public Speaking, Scientific Debates, Technical Writing, Badminton, Piano, Formula 1.